

City evidence and independent computation

SOURCE EVIDENCE

Roads, boundaries, zones
Network geography records

People and activity
Households, population, proxies

Transit and walking
GTFS, fares, service and access

Observation records
GPS, trajectories, detectors

MODEL REPRESENTATION AND UPSTREAM PREPARATION

GMNS representation
Directed physical links
Hierarchy, centroids, access
Source IDs, units and turns

City demand preparation
Statistics and boundaries
Separate activity attributes
Version / geography checks

Declared vehicle OD
Supplied demand may enter static assignment directly
No demographic compiler

Observation association
Match/associate links and service
Optional input or diagnostics
Not automatic OD or calibration

FOUR-STAGE TRAVEL-DEMAND WORKFLOW

01 / Trip generation
Activities → productions and attractions

→

02 / Trip distribution
Margins + impedance
→ person origin–destination

→

03 / Mode choice
Costs + declared choice model
→ mode-specific demand

04 / Traffic assignment
Vehicle demand → network flows

FOUR INTERNAL ASSIGNMENT LAYERS

A Native assignment

B Decomposition / distributed

C Spatial hierarchy / representation

D Coordination / verification

TWO INDEPENDENT MATHEMATICAL CONTRACTS

Static road assignment
BPR / Beckmann; period-specific vehicle OD
Frank-Wolfe · official tap-b Algorithm B; adapters vary
Finite-path reference · native L3 diagnostics
Physical-link flows and original-space checks

Finite time-expanded optimization
Separately selected graph, OD, time and capacities
Fixed arc costs · shared hard capacity
Arc-flow LP · two-phase CG · Lagrangian · ADMM
Movement-flow reconstruction and feasibility checks

SAVED OUTPUTS AND INDEPENDENT CHECKS

Saved outputs and independent checks
Zonal / mode demand · generated columns · physical-link flows · CSV / SQLite · figures
Units, conservation, objective / gap and provenance checks are specific to each instance.

CITY CASES · EVIDENCE INSTANCES

Boston
City demand and static scales
Separate bounded finite cases

Sioux Falls
Supplied-OD static benchmark
Selected 200 / 250-OD finite cases

Hong Kong
Bounded turn-aware demand
Separate static and finite cases

TOOLS AND DOCUMENTATION

Data and GMNS tools
Catalog · exports · queries

Generic RC5 engine
0.3.0-rc5 scope only

Versioned method code
Algorithms · case adapters

Examples and checks
Saved results · validation
Independent validators